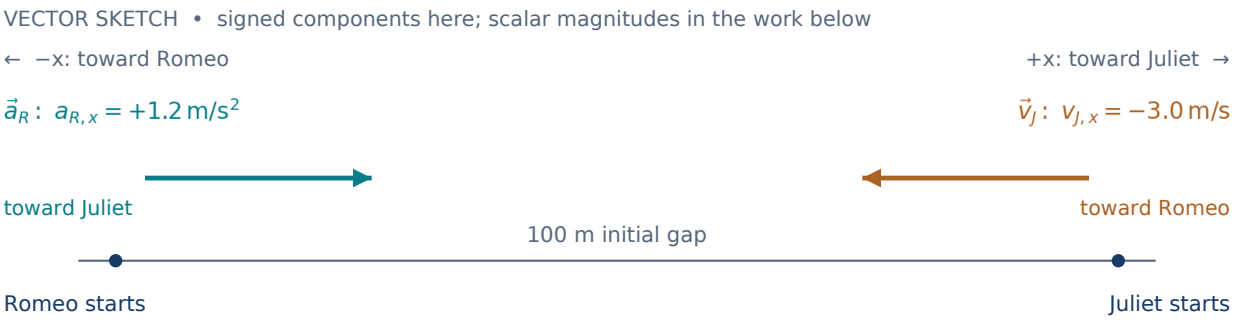
# Example 1: Romeo and Juliet

**Problem.** Romeo starts from rest under a tree, 100 m from Juliet. Juliet runs toward him at 3.0 m/s while Romeo accelerates toward her at 1.2 m/s². Answer (a)–(e): compare their times, link their distances, write each motion equation, and find the meeting time.




**REFERENCE AND CLOCK**

Let t start when both move. Measure each person's distance toward the other as a positive scalar.

**WHAT TO FIND**

Find the time when their two travel distances fill the 100 m gap.

### 1. Connect their distances and times

The arrows in the sketch show opposite directions. In the calculation, d is distance along each person's own path, so both distances are positive and add to the gap. They start together and meet together.

$$t_R = t_J = t$$
$$d_R + d_J = 100\,\text{m}$$

### 2. Use scalar magnitudes in both motion equations

Use 3.0 m/s for Juliet's speed and 1.2 m/s² for Romeo's acceleration magnitude. Romeo starts from rest; all quantities in these distance equations are scalars.

$$d_J = v_J t = (3.0\,\text{m/s})t$$
$$v_{i,R} = 0, \qquad d_R = v_{i,R}t + \tfrac{1}{2}a_R t^2$$
$$d_R = \tfrac{1}{2}(1.2\,\text{m/s}^2)t^2 = 0.60t^2$$

### 3. Add the distances and solve

Use t in seconds and distances in metres. The two expressions must total 100 m; keep the positive quadratic root.

$$0.60t^2 + 3.0t = 100$$
$$t = \frac{-3.0 + \sqrt{3.0^2 + 4(0.60)(100)}}{1.20} = 10.6498\,\text{s}$$

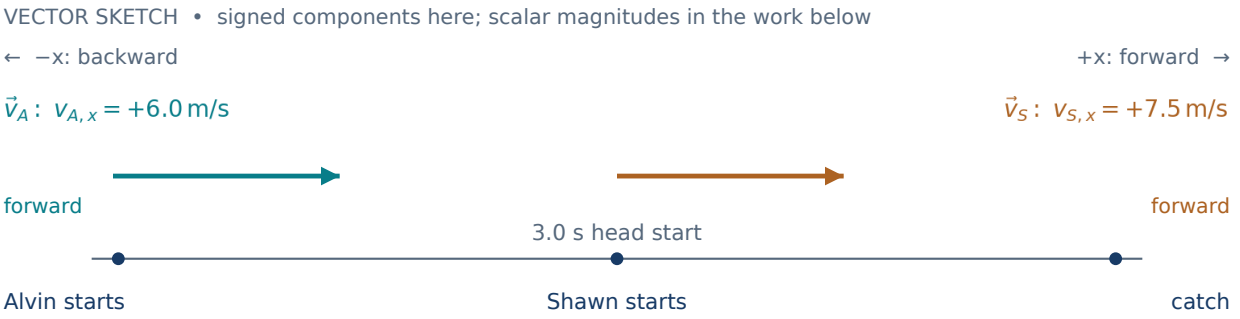
### 4. Check the meeting

Romeo runs 68.05 m and Juliet runs 31.95 m. Their scalar distances add to 100 m.

**FINAL They meet 10.6 s after they start.**

# Example 2: Shawn catches Alvin

**Problem.** Alvin starts 3.0 s before Shawn and runs at 6.0 m/s. Shawn runs at 7.5 m/s. Answer (a)–(d): compare distances and times, write the motion equations, and determine how long Shawn runs before catching Alvin.




**REFERENCE AND CLOCK**

Measure each runner's positive distance from their shared start line. Let t be Shawn's running time.

**WHAT TO FIND**

Find t; then distinguish Shawn's clock from Alvin's clock.

### 1. Relate the clocks

Alvin starts 3.0 s earlier, so by the time Shawn has run for t seconds, Alvin has run for t + 3.0 s.

$$t_A = t + 3.0\,\text{s}$$

### 2. Write both travel distances

They start from the same mark and meet at one point. Each scalar distance is speed multiplied by that runner's own elapsed time.

$$d_S = (7.5\,\text{m/s})t$$
$$d_A = (6.0\,\text{m/s})(t + 3.0\,\text{s})$$

### 3. Set the distances equal and solve

Equate the distances, expand, and isolate t.

$$7.5t = 6.0(t + 3.0)$$
$$1.5t = 18 \quad \Longrightarrow \quad t = 12\,\text{s}$$

### 4. Check both clocks and distances

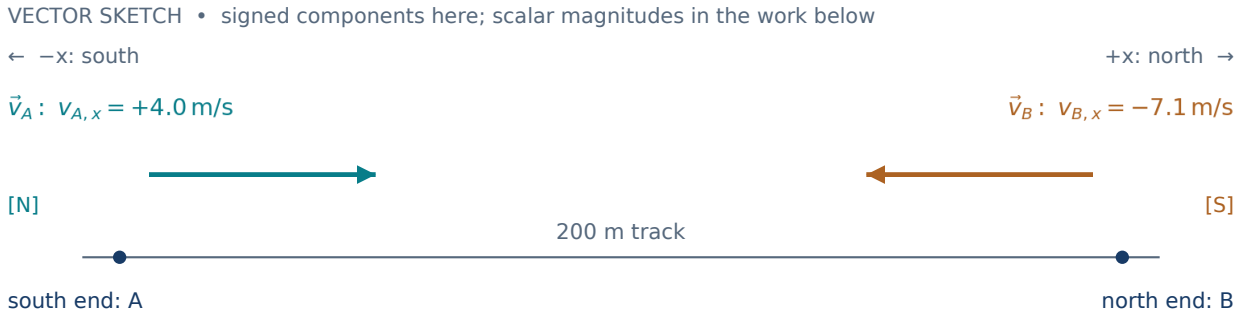
Shawn has run 12 s, Alvin 15 s. Both have traveled 90 m.

$$7.5(12) = 6.0(15) = 90\,\text{m}$$

**FINAL Shawn catches Alvin after 12 s of Shawn's running, 15 s after Alvin starts, at 90 m.**

# 1. Runners at opposite ends

**Problem.** Runner A leaves the south end of a 200. m north–south track at 4.0 m/s [N]. Runner B leaves the north end toward the south. They meet after 18 s. Find B's velocity.




**REFERENCE AND CLOCK**

Both run for 18 s. Use positive travel distances along each runner’s path.

**WHAT TO FIND**

Find runner B’s velocity, including its direction.

### 1. Identify the link

They move toward each other at the same time; the distances they cover add to the track length.

$$d_A + d_B = 200\,\text{m}$$

### 2. Find A’s distance

At constant speed, distance equals speed times time.

$$d_A = (4.0\,\text{m/s})(18\,\text{s}) = 72\,\text{m}$$

### 3. Find B’s distance and velocity

B covers the rest of the track in the same 18 s. Its direction is south.

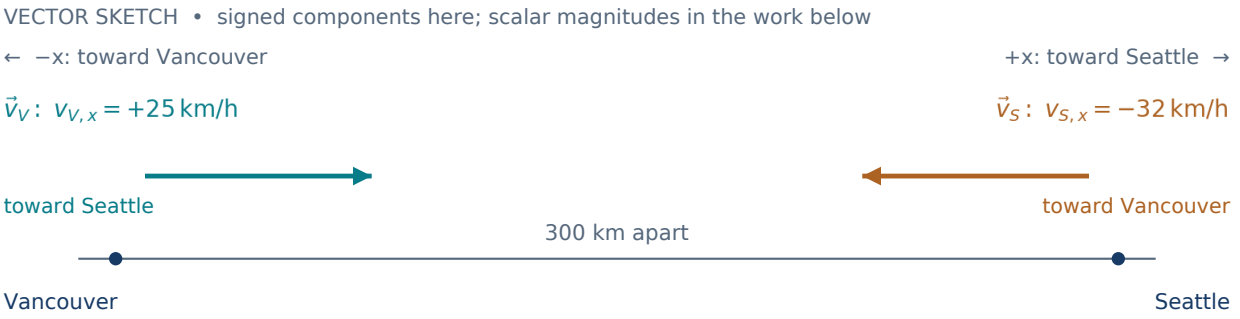
$$d_B = 200 - 72 = 128\,\text{m}$$

$$v_B = \frac{128\,\text{m}}{18\,\text{s}} = 7.11\,\text{m/s}$$

**FINAL Runner B moves at 7.1 m/s [S].**

# 2. Vancouver–Seattle cyclists

**Problem.** The cities are about 300 km apart. A cyclist leaves Vancouver toward Seattle at 25 km/h; a second cyclist leaves Seattle toward Vancouver 2.0 h later at 32 km/h. How far from Vancouver do they meet?




**REFERENCE AND CLOCK**

Let t be hours after the Vancouver departure. The Seattle cyclist rides for t − 2.0 h.

**WHAT TO FIND**

Locate the meeting point from Vancouver.

### 1. Relate the times

The second cyclist starts 2.0 h later. Both positions are compared at the same meeting instant.

$$t_V = t, \qquad t_S = t - 2.0\,\text{h}$$

### 2. Fill the distance between the cities

Each cyclist's distance is speed times that cyclist's travel time. Their distances toward one another sum to 300 km.

$$25t + 32(t - 2.0) = 300$$

### 3. Find the departure-to-meeting time

Expand and solve. The result is later than 2.0 h, as required for the Seattle cyclist to have started.

$$57t - 64 = 300$$
$$t = \frac{364}{57} = 6.38596\,\text{h}$$

### 4. Measure from the requested city

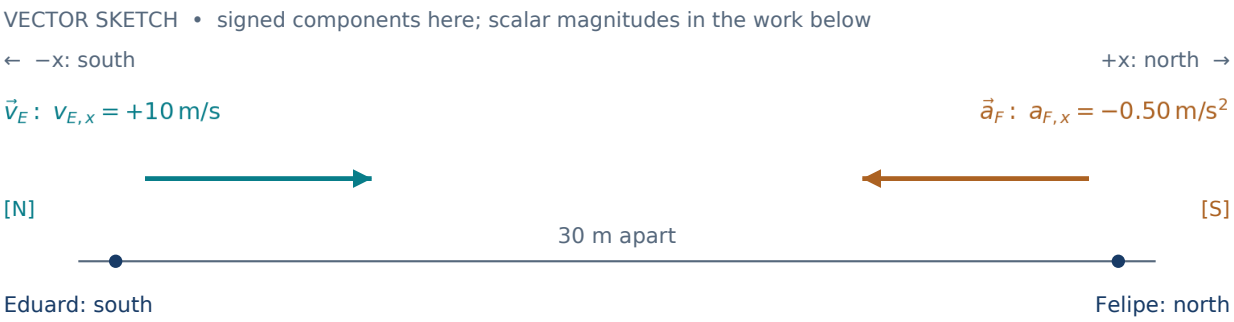
The Vancouver cyclist's distance gives the meeting location. The other cyclist travels 140.35 km; together these give 300 km.

$$d_V = (25)(6.38596) = 159.65\,\text{km}$$

**FINAL They meet about 160 km from Vancouver.**

# 3. Eduard and Felipe

**Problem.** Eduard runs at 10. m/s [N] while 30. m from Felipe. Felipe starts from rest and accelerates at 0.50 m/s² [S] toward Eduard. When does Felipe tackle Eduard?




**REFERENCE AND CLOCK**

Eduard must start south of Felipe: northward and southward motion then close the gap.

**WHAT TO FIND**

Find their common travel time until contact.

### 1. Express the meeting link

They begin together and move toward each other, so their positive travel distances add to 30 m.

$$d_E + d_F = 30\,\text{m}$$

### 2. Model both distances

Eduard's speed is constant. Felipe starts from rest, so the initial-velocity term in his displacement is zero.

$$d_E = (10\,\text{m/s})t$$

$$d_F = \tfrac{1}{2}(0.50\,\text{m/s}^2)t^2 = 0.25t^2$$

### 3. Solve the quadratic

Keep the positive time and discard the root before the start.

$$0.25t^2 + 10t - 30 = 0$$

$$t = \frac{-10 + \sqrt{10^2 + 4(0.25)(30)}}{0.50} = 2.8035\,\text{s}$$

### 4. Check the gap

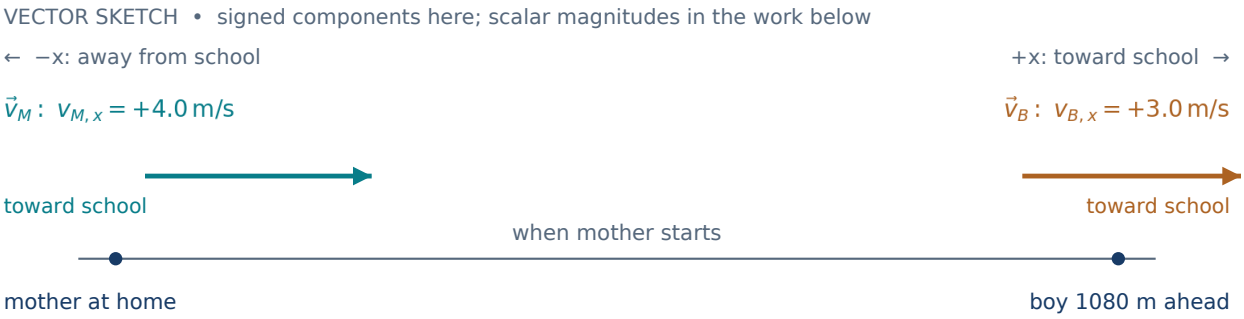
Eduard covers 28.04 m and Felipe 1.96 m, together 30.00 m.

$$10(2.8035) + 0.25(2.8035)^2 \approx 30\,\text{m}$$

**FINAL Felipe tackles Eduard after 2.8 s.**

# 4. The forgotten lunch

**Problem.** A boy runs toward school at his usual pace of 15 m in 5.0 s. His mother leaves home 360 s later and runs after him at 4.0 m/s. Find (a) his distance at her departure, (b) her catch time, and (c) his distance from home then.




**REFERENCE AND CLOCK**

Measure positive distance from home. Let t begin when the mother starts.

**WHAT TO FIND**

Find the boy's head start, the catch time, and the catch position.

### 1. Find the boy's speed and initial lead (a)

Convert his normal pace to speed. In the 360 s before his mother leaves, he travels 1080 m from home.

$$v_B = \frac{15\,\text{m}}{5.0\,\text{s}} = 3.0\,\text{m/s}$$
$$d_{B,0} = (3.0)(360) = 1080\,\text{m}$$

### 2. Write distances on the mother's clock

After she begins, the mother starts at home; the boy has already traveled 1080 m and keeps moving.

$$d_M = (4.0)t$$
$$d_B = 1080 + (3.0)t$$

### 3. Set equal distances from home (b)

At a catch they are the same distance from home, so the mother must erase the 1080 m lead at a relative speed of 1.0 m/s.

$$4.0t = 1080 + 3.0t \quad \Longrightarrow \quad t = 1080\,\text{s}$$

### 4. Find the shared distance from home (c)

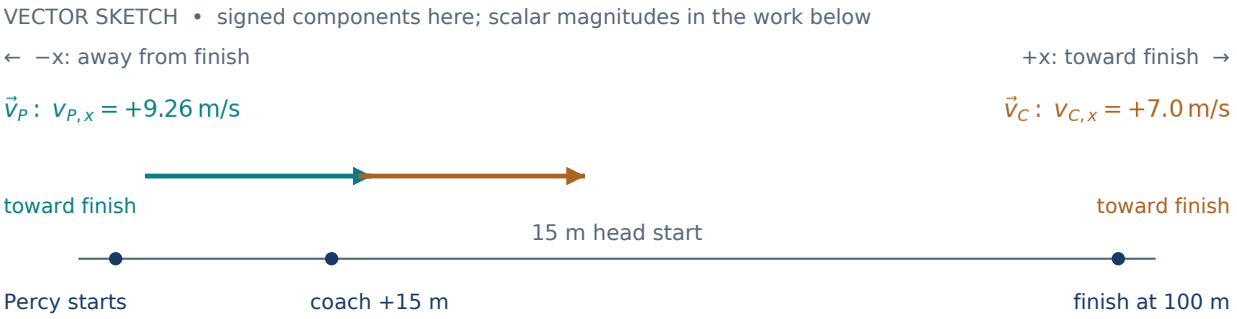
Use either person's distance equation at that time.

$$d = (4.0)(1080) = 4320\,\text{m}$$

**FINAL (a) 1080 m; (b) 1080 s after the mother starts; (c) 4320 m from home.**

# 5. Percy Williams and his coach

**Problem.** Percy runs 100. m in 10.8 s; his coach averages 7.0 m/s. (a) With a 15 m distance head start for the coach, find when and where Percy catches him. (b) What distance head start lets them finish 100 m at the same time?




**REFERENCE AND CLOCK**

Measure distance from Percy's line. The coach begins 15 m from that line in (a).

**WHAT TO FIND**

Find the catch time and distance from Percy's line, then a head start for a tie.

### 1. Find Percy's average speed

Both runners are modeled with constant average speeds.

$$v_P = \frac{100\,\text{m}}{10.8\,\text{s}} = 9.25926\,\text{m/s}$$

### 2. Equate distances from Percy's line (a)

The coach begins 15 m ahead. Percy catches him when both are the same distance from Percy's line.

$$d_P = v_P t, \qquad d_C = 15 + 7.0t$$
$$9.25926t = 15 + 7.0t$$

### 3. Solve and state which distance is which

The catch happens 61.48 m from Percy's line. The coach runs only 46.48 m because he starts at 15 m.

$$t = \frac{15}{9.25926 - 7.0} = 6.63934\,\text{s}$$
$$d_P = 9.25926(6.63934) = 61.48\,\text{m}$$

### 4. Arrange an equal finish time (b)

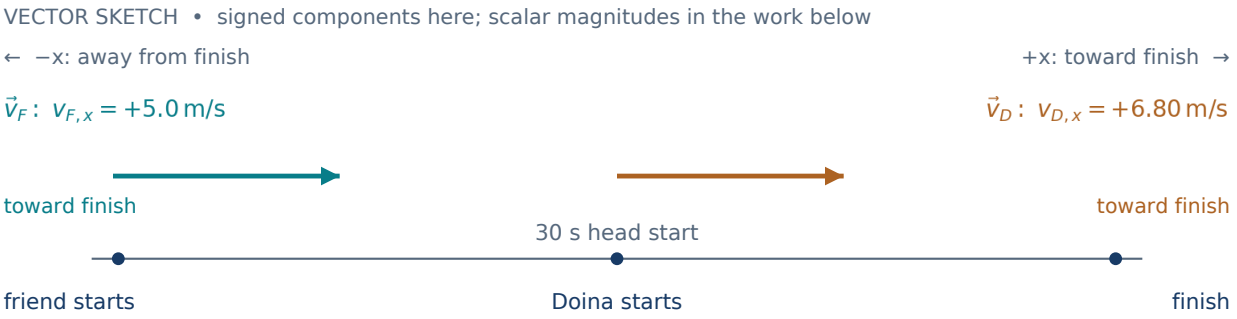
Percy takes 10.8 s. The coach can run 75.6 m in 10.8 s, so he needs the remaining distance as his head start.

$$d_{\text{head}} = 100 - (7.0)(10.8) = 24.4\,\text{m}$$

**FINAL (a) 6.6 s; at 61 m from Percy's line (46 m beyond the coach's mark). (b) 24.4 m head start.**

# 6. Doina and her friend

**Problem.** Doina runs 800 m in 117.6 s; her friend runs at 5.0 m/s and starts 30. s earlier. Find (a) the friend's lead when Doina starts, (b) the catch time, (c) distance remaining, and (d) the head start needed for a tie at the finish.




**REFERENCE AND CLOCK**

Measure positive distance from the race start. Let t start when Doina runs; her friend started 30 s earlier.

**WHAT TO FIND**

Answer (a)–(d) using distances from one start line.

### 1. Calculate speeds and the existing lead (a)

The friend covers 5.0 m each second; 30 s before Doina starts gives a 150 m lead.

$$v_D = \frac{800}{117.6} = 6.80272\,\text{m/s}$$
$$d_0 = (5.0)(30) = 150\,\text{m}$$

### 2. Find the catch time (b)

At the catch, Doina's distance equals the friend's 150 m lead plus the distance the friend runs during t.

$$(6.80272)t = 150 + (5.0)t$$
$$t = \frac{150}{6.80272 - 5.0} = 83.2075\,\text{s}$$

### 3. Find what remains of the race (c)

At the catch, Doina has run 566.04 m along the 800 m race.

$$d_{\text{remaining}} = 800 - (6.80272)(83.2075) = 233.96\,\text{m}$$

### 4. Choose a head start that creates a tie (d)

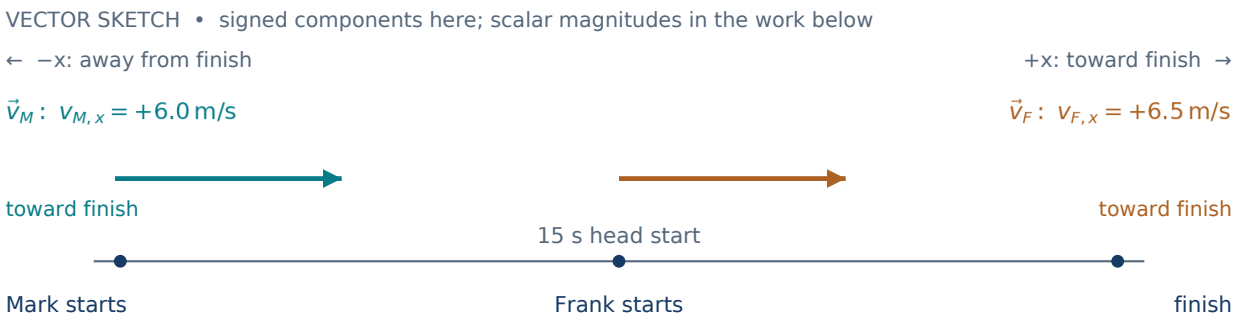
The friend needs 160 s for 800 m; Doina needs 117.6 s. Their difference is the friend's required time head start, corresponding to 212 m.

$$\Delta t = \frac{800}{5.0} - 117.6 = 42.4\,\text{s}$$
$$d_{\text{head}} = (5.0)(42.4) = 212\,\text{m}$$

**FINAL (a) 150 m; (b) 83 s after Doina starts; (c) 234 m left; (d) 42.4 s or 212 m head start.**

# 7. Mark and Frank on foot

**Problem.** Mark covers 12 m in 2.0 s and starts first. Frank ties his shoelace for 15 s before setting off at 6.5 m/s. Find (a) Frank's catch time and (b) who finishes a 3 km race first, and by what distance.




**REFERENCE AND CLOCK**

Measure distance from the race start. Let t be seconds after Frank starts; Mark began 15 s earlier.

**WHAT TO FIND**

Find Frank’s catch time and Mark’s distance when Frank finishes 3 km.

### 1. Find the speed and head start

Mark’s speed is 6.0 m/s. By Frank’s departure Mark is 90 m ahead.

$$v_M = \frac{12\,\text{m}}{2.0\,\text{s}} = 6.0\,\text{m/s}$$
$$d_0 = (6.0)(15) = 90\,\text{m}$$

### 2. Set the runners’ distances equal (a)

Using the clock that begins when Frank runs, Mark starts 90 m ahead. Frank closes that lead at 0.5 m/s.

$$d_F = 6.5t, \qquad d_M = 90 + 6.0t$$
$$6.5t = 90 + 6.0t \quad \Longrightarrow \quad t = 180\,\text{s}$$

### 3. Find when Frank reaches 3 km

Frank’s travel time is 3000/6.5 s; Mark has spent an additional 15 s running by that instant.

$$t_{F,\,\text{finish}} = \frac{3000}{6.5} = 461.538\,\text{s}$$
$$t_{M,\,\text{then}} = 15 + 461.538 = 476.538\,\text{s}$$

### 4. Compare distances at the same instant (b)

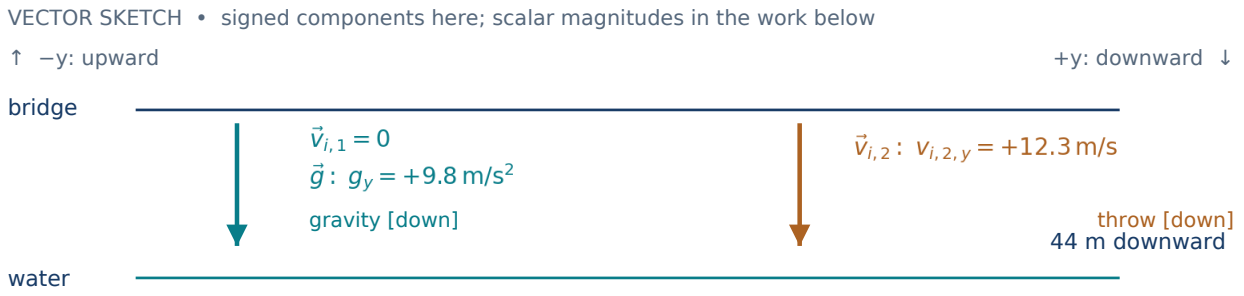
When Frank crosses the finish, Mark has traveled 2859.23 m, leaving a separation of 140.77 m.

$$d_M = (6.0)(476.538) = 2859.23\,\text{m}$$
$$3000 - 2859.23 = 140.77\,\text{m}$$

**FINAL (a) 180 s after Frank starts; (b) Frank wins by about 140 m at the instant he finishes.**

# 8. Two stones from a bridge

**Problem.** A stone is dropped from a bridge 44 m above the water. Another is thrown downward 1.0 s later. They hit the water together. Find the second stone's initial velocity. Neglect air resistance.




**REFERENCE AND CLOCK**

Measure positive distance downward from the bridge. Stone 1 starts at t = 0; stone 2 starts 1.0 s later.

**WHAT TO FIND**

Find stone 2's initial speed; its direction is downward in the sketch.

### 1. Find the first stone's fall time

Stone 1 starts from rest and falls 44 m with gravitational acceleration 9.8 m/s².

$$44 = \frac{1}{2}(9.8)t_1^2$$

$$t_1 = \sqrt{\frac{88}{9.8}} = 2.9966\,\text{s}$$

### 2. Relate the flight times

Both stones land together, but the second starts 1.0 s later.

$$t_2 = t_1 - 1.0 = 1.9966\,\text{s}$$

### 3. Apply the second stone's distance equation

Unlike stone 1, stone 2 has a nonzero initial speed. The speed and gravitational acceleration magnitude are positive scalars because both act along the downward path.

$$44 = v_{i,2}t_2 + \frac{1}{2}(9.8)t_2^2$$

$$v_{i,2} = \frac{44 - \frac{1}{2}(9.8)(1.9966)^2}{1.9966} = 12.25\,\text{m/s}$$

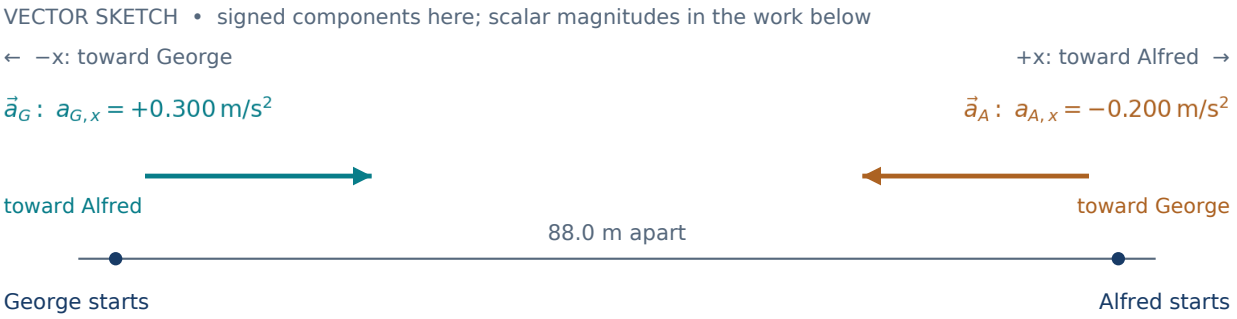
### 4. Interpret the worksheet's rounded value

Using rounded intermediate times of 3.0 s and 2.0 s gives 12.2 m/s. Keeping calculator precision gives 12.25 m/s before the final rounding.

**FINAL About 12.3 m/s [down]; the worksheet's 12.2 m/s uses early rounding.**

# 9. Knights riding toward each other

**Problem.** Sir George and Sir Alfred start from rest 88.0 m apart and ride directly toward one another. Their accelerations are 0.300 m/s² and 0.200 m/s², respectively. Find the collision time and place relative to George's starting point.




**REFERENCE AND CLOCK**
Measure each knight's positive travel distance toward the other from his own start.

**WHAT TO FIND**
Find the shared collision time and George's distance to the collision.

### 1. State the meeting condition

They start simultaneously from rest. Their distances toward each other fill the 88.0 m gap.

$$d_G + d_A = 88.0\,\text{m}$$

### 2. Write each constant-acceleration distance

Both initial speeds are zero, so each distance is one-half acceleration times time squared.

$$d_G = \tfrac{1}{2}(0.300)t^2$$

$$d_A = \tfrac{1}{2}(0.200)t^2$$

### 3. Find the time

Substitute both distances into the gap condition.

$$0.150t^2 + 0.100t^2 = 88.0$$

$$t = \sqrt{\frac{88.0}{0.250}} = 18.7617\,\text{s}$$

### 4. Locate the collision

George's travel distance is the requested distance from his start. Alfred covers the remaining 35.2 m.

$$d_G = 0.150(18.7617)^2 = 52.8\,\text{m}$$

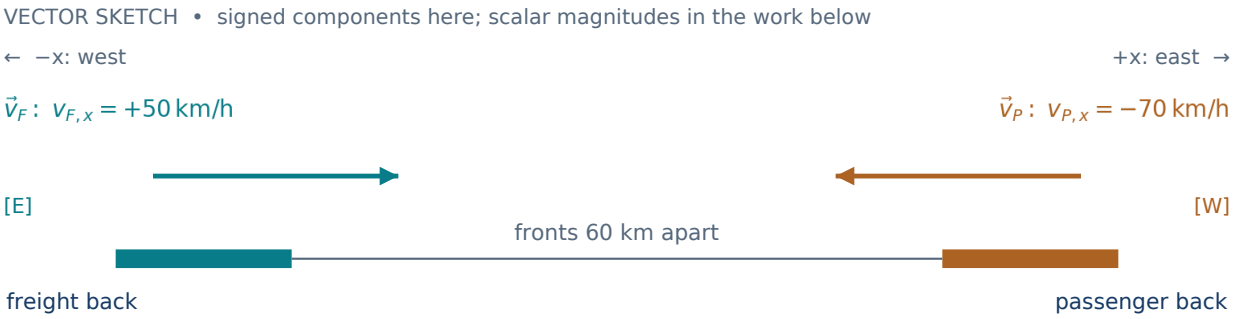
$$52.8 + 35.2 = 88.0\,\text{m}$$

**Rounding check:** The worksheet's 53.0 m results from substituting its rounded 18.8 s; use 52.8 m with unrounded time.

**FINAL They collide after 18.8 s, 52.8 m from George's starting point toward Alfred.**

# 10. Two trains passing

**Problem.** A 1.4 km freight train travels 50. km/h [E] and a 1.0 km passenger train travels 70. km/h [W]. They depart from cities 60 km apart, moving toward one another on parallel tracks. Find (a) when the back ends pass, (b) where that happens, and (c) the duration of overlap.




**REFERENCE AND CLOCK**

The train fronts start 60 km apart. Use positive closing distances and a shared departure clock.

**WHAT TO FIND**

Find last contact, the freight back's travel distance, and total overlap time.

### 1. Draw the whole trains

Their back ends start farther apart than their fronts: add both train lengths to the 60 km front-to-front gap.

$$d_{\text{backs}} = 60 + 1.4 + 1.0 = 62.4\ \text{km}$$

### 2. Find when the back ends pass (a)

The trains approach at the sum of their speeds, 120 km/h. Back-to-back contact marks the end of passing.

$$v_{\text{closing}} = 50 + 70 = 120\ \text{km/h}$$

$$t_{\text{last}} = \frac{62.4}{120} = 0.520\ \text{h}$$

### 3. Find how far the freight back travels (b)

Multiply the freight train's scalar speed by the time to last contact. The sketch supplies its eastward direction.

$$d_{F,\,\text{back}} = (50)(0.520) = 26.0\ \text{km}$$

### 4. Subtract first contact from last contact (c)

The fronts first meet after closing the 60 km initial gap. The extra 2.4 km corresponds to the total time of overlap.

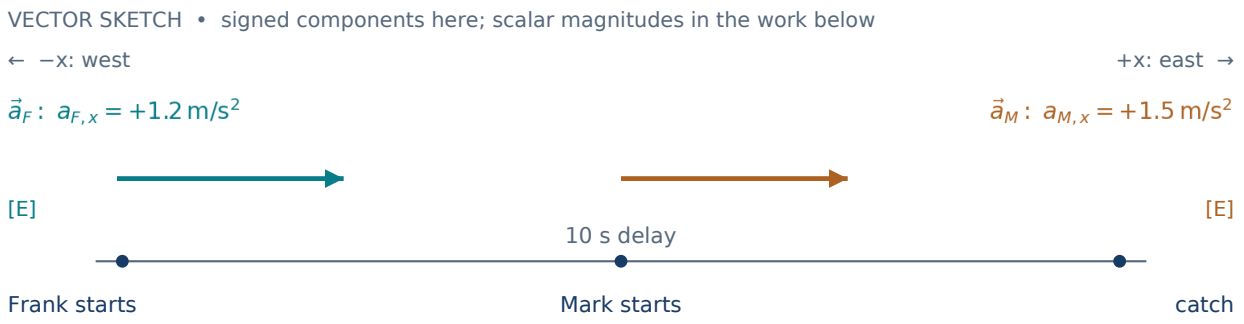
$$t_{\text{first}} = \frac{60}{120} = 0.500\ \text{h}$$

$$\Delta t = 0.520 - 0.500 = 0.020\ \text{h} = 72\ \text{s}$$

**FINAL (a) 0.52 h; (b) 26 km [E] from the freight back's start; (c) 0.020 h = 72 s overlapping.**

# 11. Frank and Mark on bikes

**Problem.** Frank starts from rest and accelerates east at 1.2 m/s². Mark starts from the same line 10 s later and accelerates at 1.5 m/s² [E]. Where and when does Mark catch Frank?




**REFERENCE AND CLOCK**

Measure eastward distance from their shared start. Let t begin when Frank starts; Mark begins at t = 10 s.

**WHAT TO FIND**

Find the catch distance and time, with t at least 10 s.

### 1. Write the two running times

Frank has accelerated for t seconds; Mark has accelerated for t − 10 seconds.

$$t_F = t, \qquad t_M = t - 10\,\text{s}, \quad t \geq 10\,\text{s}$$

### 2. Use the scalar distance equation from rest

Both start from the same line and accelerate east. Their positive travel distances must match at the catch.

$$d_F = \tfrac{1}{2}(1.2)t^2 = 0.60t^2$$

$$d_M = \tfrac{1}{2}(1.5)(t-10)^2 = 0.75(t-10)^2$$

### 3. Solve and reject the invalid root

Equating distances gives two algebraic roots. The 5.28 s root is before Mark has started and is outside his equation's time interval.

$$0.60t^2 = 0.75(t-10)^2$$

$$t^2 - 100t + 500 = 0$$

$$t = 5.2786\,\text{s} \quad \text{or} \quad 94.7214\,\text{s}$$

### 4. Find the travel distance

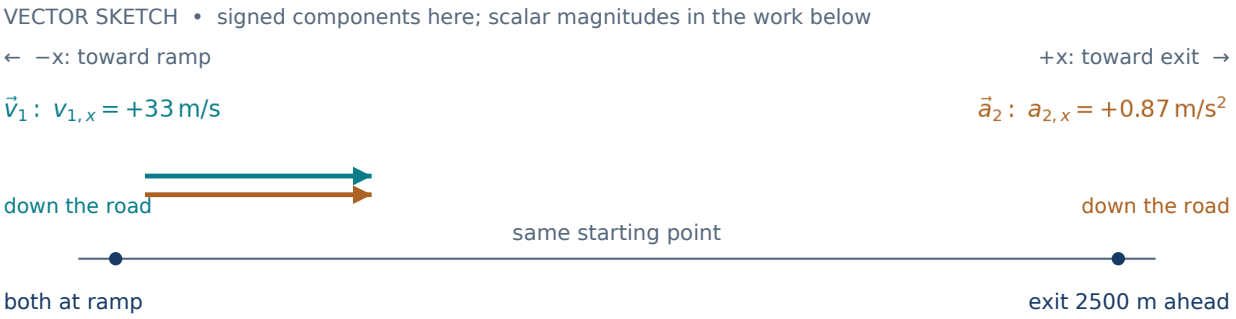
Use Frank's distance at the valid later time; Mark has actually cycled for 84.72 s.

$$d = 0.60(94.7214)^2 = 5383.28\,\text{m} = 5.38328\,\text{km}$$

**FINAL Mark catches Frank 5.38 km [E] from the start, 94.7 s after Frank starts.**

# 12. Two cars meet at the exit

**Problem.** A car passes a ramp at a constant 33 m/s. At that instant a second car joins from the ramp at rest and accelerates uniformly. What acceleration makes them meet for the first time at the exit 2.5 km ahead?




**REFERENCE AND CLOCK**

Measure positive distance along the road from the ramp to the exit, 2500 m away. Both start at t = 0.

**WHAT TO FIND**

Find the second car's constant acceleration for their next meeting at the exit.

### 1. Find the first car's travel time

The first car moves at constant speed for 2500 m.

$$t = \frac{2500\,\text{m}}{33\,\text{m/s}} = 75.7576\,\text{s}$$

### 2. Make the second car reach the same exit then

The second car starts from rest; apply the constant-acceleration displacement equation.

$$2500 = \frac{1}{2}a(75.7576)^2$$

$$a = \frac{2(2500)}{(75.7576)^2} = 0.8712\,\text{m/s}^2$$

### 3. Check that this is the first later meeting

Equating their travel distances gives the shared ramp position at t = 0 and exactly one positive time after it.

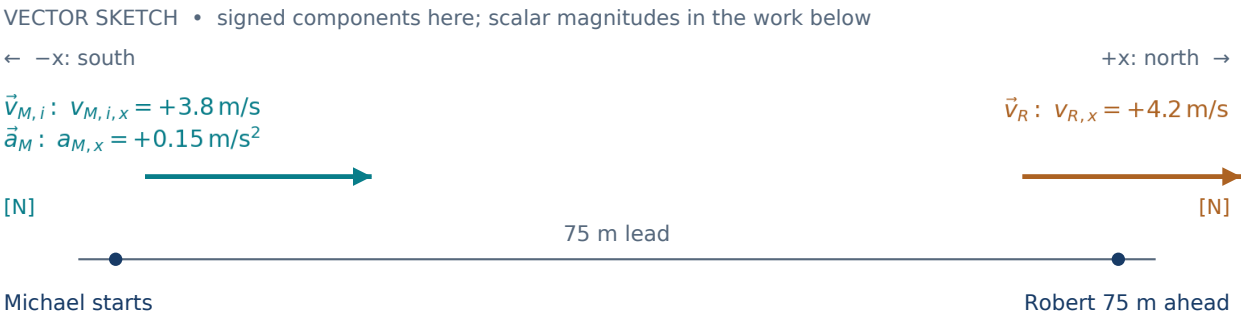
$$33t = \frac{1}{2}at^2$$

$$t = 0 \quad \text{or} \quad t = \frac{66}{a} = 75.7576\,\text{s}$$

**FINAL The second car must accelerate at 0.87 m/s² in the direction of travel.**

# 13. Michael catches Robert

**Problem.** Michael runs at 3.8 m/s [N] and is 75 m behind Robert, who maintains 4.2 m/s [N]. Michael accelerates at 0.15 m/s² [N]. How long until he catches Robert?




**REFERENCE AND CLOCK**

Measure positive distance north from Michael's start. Robert begins 75 m farther along the route.

**WHAT TO FIND**

Find the time when both have traveled to the same place.

### 1. Write both scalar distance equations

Robert begins 75 m ahead at constant speed. Michael starts slower but accelerates northward.

$$d_M = 3.8t + \frac{1}{2}(0.15)t^2$$
$$d_R = 75 + 4.2t$$

### 2. Apply the catch condition

Set the total distances measured from Michael's start equal and collect terms.

$$3.8t + 0.075t^2 = 75 + 4.2t$$
$$0.075t^2 - 0.40t - 75 = 0$$

### 3. Solve for the physical time

The positive root is the catch; the negative root refers to before the stated start.

$$t = \frac{0.40 + \sqrt{0.40^2 + 4(0.075)(75)}}{0.150} = 34.4017\,\text{s}$$

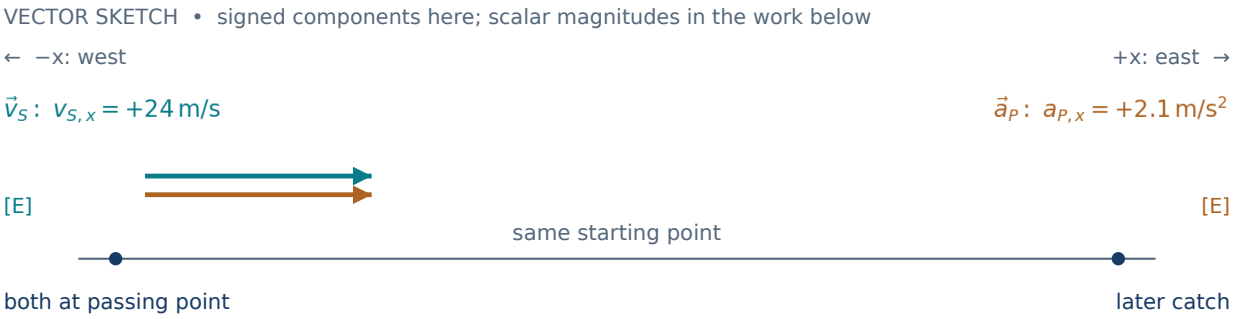
### 4. Interpret the result

Michael is initially 0.40 m/s slower, so the gap first widens. His acceleration later raises his speed enough to overtake Robert.

**FINAL Michael catches Robert after about 34 s (34.4 s before rounding).**

# 14. Police car catches a speeder

**Problem.** A car travels east at a constant 24 m/s and passes a stationary police car. At that instant the police car accelerates east from rest at 2.1 m/s². Find (a) the catch time and (b) the distance traveled by each car.




**REFERENCE AND CLOCK**

Measure positive travel distance east from the passing point. Start the clock as the speeder passes.

**WHAT TO FIND**

Find the next meeting time and their shared distance from that point.

### 1. Model both distances

The speeder has constant speed; the police car starts at rest with a positive acceleration magnitude eastward.

$$d_S = (24)t$$
$$d_P = \tfrac{1}{2}(2.1)t^2 = 1.05t^2$$

### 2. Find the later shared time (a)

Equating distances also gives t = 0, their initial passing. Choose the positive later root.

$$24t = 1.05t^2$$
$$t(1.05t - 24) = 0$$
$$t = \frac{24}{1.05} = 22.8571\,\text{s}$$

### 3. Find the shared distance (b)

Substitute the positive root into either distance equation; both yield the same value.

$$d = (24)(22.8571) = 548.571\,\text{m}$$
$$d = 1.05(22.8571)^2 = 548.571\,\text{m}$$

**FINAL (a) About 23 s after the speeder passes; (b) about 549 m east of the passing point.**